

Supervised Learning, COMP0078
Sample Examination Paper 4 (Level 2)
Suitable for Cohorts: 25/26

Answer ALL EIGHT questions.

Notation: Let $[m] := \{1, \dots, m\}$. We also overload notation so that

$$[\text{pred}] := \begin{cases} 1 & \text{pred is true} \\ 0 & \text{pred is false} \end{cases}$$

Marks for each part of each question are indicated in square brackets.
Calculators are NOT permitted.

1.

- (a) State the No Free Lunch theorem for supervised learning. What does it imply about the necessity of prior knowledge (hypothesis class) for learning? [5 marks]
- (b) The curse of dimensionality affects k -NN. Explain why: in high dimensions, even “nearby” points can be far apart, and the volume of a ball of radius r relative to a ball of radius R shrinks as $(r/R)^d$. What does this mean for the convergence rate of k -NN? [5 marks]

2.

- (a) A problem is *well-posed* (Hadamard) if: (1) a solution exists, (2) the solution is unique, (3) the solution depends continuously on the data.
Why is the least squares problem $\min_{\mathbf{w}} \|X\mathbf{w} - \mathbf{y}\|^2$ potentially ill-posed? How does ridge regression (adding $\lambda\|\mathbf{w}\|^2$) fix this? [5 marks]
- (b) Is the following statement true or false? “If f is convex and g is convex, then $f(g(\cdot))$ is convex.”
Provide a counterexample or proof. Compare with the case when g is affine. [10 marks]

3.

- (a) For the hard-margin SVM, we use the *canonical parameterization* where $\min_i y_i(\mathbf{w}^\top \mathbf{x}_i + b) = 1$.

Explain why this choice is made. Under this parameterization, what is the geometric margin in terms of $\|\mathbf{w}\|$? [5 marks]

- (b) Starting from the hard-margin SVM primal:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1$$

Write the Lagrangian $L(\mathbf{w}, b; \alpha)$. By setting $\frac{\partial L}{\partial \mathbf{w}} = 0$ and $\frac{\partial L}{\partial b} = 0$, derive the two key conditions that lead to the dual. [5 marks]

4.

- (a) In cost-complexity pruning, we minimize:

$$C_\lambda(T) = \sum_{p=1}^{|T|} m_p Q_p(T) + \lambda |T|$$

The weakest link pruning algorithm successively collapses nodes. Describe the criterion for choosing which node to collapse at each step. [5 marks]

- (b) Using the Chernoff bound: if $2T + 1$ independent voters each have accuracy $\frac{1}{2} + \gamma$, the probability the majority is wrong is at most $\exp(-2\gamma^2 T)$.

What does this result say about ensemble methods? Why do Random Forests try to make trees “independent”? [5 marks]

5.

- (a) The Weighted Average algorithm for entropic loss $L_{en}(y, \hat{y}) = y \ln(y/\hat{y}) + (1 - y) \ln((1 - y)/(1 - \hat{y}))$ uses a key identity involving KL divergence:

$$L_{en}(y_t, \hat{y}_t) - \sum_i u_i L_{en}(y_t, x_{t,i}) = d(\mathbf{u}, \mathbf{v}_t) - d(\mathbf{u}, \mathbf{v}_{t+1})$$

Explain intuitively why this “telescoping” property leads to a regret bound of $\ln n$ (the initial KL divergence from uniform). [5 marks]

- (b) The Variable Share algorithm modifies Fixed Share by making the share amount depend on the expert’s loss. When would Variable Share outperform Fixed Share? What is the intuition? [5 marks]

6.

- (a) Distinguish between *transductive* and *inductive* semi-supervised learning. Which type does Laplacian interpolation perform? Can the harmonic solution be extended to new test points not in the original graph? [5 marks]
- (b) The harmonic solution can be interpreted via random walks: $\bar{u}_i$ equals the probability that a random walk from vertex i reaches the +1 labeled vertices before the -1 labeled vertices.

For a path graph $v_1 - v_2 - v_3$ with v_1 labeled +1 and v_3 labeled -1, use this interpretation to find $\bar{u}_2$. [5 marks]

7.

- (a) The validation set bound states: with probability $\geq 1 - \delta$,

$$L_{\mathcal{D}}(h) \leq L_V(h) + \sqrt{\frac{\ln(1/\delta)}{2m}}$$

This follows from Hoeffding's inequality. Explain why we need a *separate* validation set V that was not used to select h . [5 marks]

- (b) For the class of halfspaces with margin γ over points with $\|\mathbf{x}\| \leq R$, the VC dimension is $O(R^2/\gamma^2)$, which can be much smaller than the ambient dimension.

Why is this significant for SVMs? How does maximizing the margin help generalization according to this result? [5 marks]

8.

- (a) In the bandit setting, we only observe $\ell_{t,\hat{y}_t}$ (loss of chosen action). EXP3 constructs the importance-weighted estimator:

$$\hat{\ell}_{t,i} = \frac{\ell_{t,i}}{v_{t,i}} \mathbf{1}[i = \hat{y}_t]$$

Show that $\mathbb{E}[\hat{\ell}_{t,i}] = \ell_{t,i}$ (unbiasedness). Why is division by $v_{t,i}$ necessary? [5 marks]

- (b) EXP3's regret bound is $O(\sqrt{nT \ln n})$ compared to Hedge's $O(\sqrt{T \ln n})$.

Explain intuitively why the bandit setting incurs an extra $\sqrt{n}$ factor. What is the "exploration vs exploitation" tradeoff in this context? [5 marks]

END OF PAPER